



BA510 VC Platform Success Predictor

By Elmir Shabayev

Meet the Team



Elmir Shabayev
Unemployed



Claude
Head Reason for
Unemployment



ChatGPT
Former Head Reason
for Unemployment

Prologue



Problem Substantiation:

VC Firms



- **Invest in Startups.**
- Have **some successes, many failures.**
- **Face pressure** to perform from their Limited Partners.



Platforms



- **Connect two parties** (Amazon, Facebook, Craigslist).
- **Network effects and minimum liability.**

Problem Substantiation:

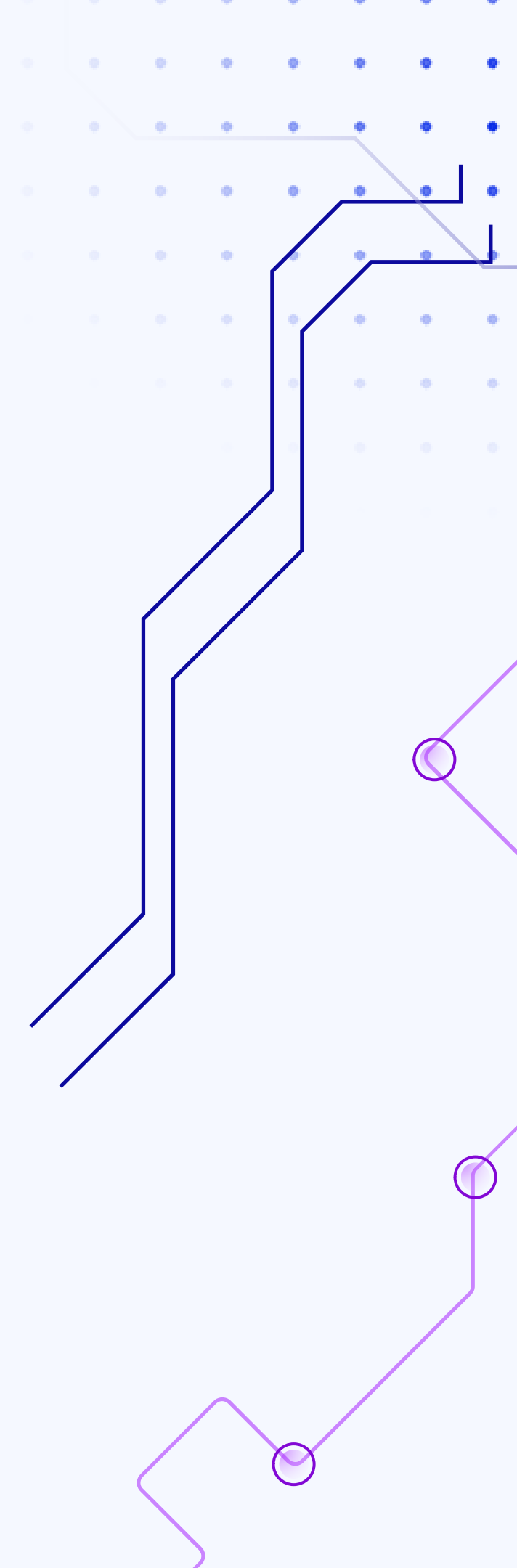
Result:

VC Firms want to be able to
predict and invest in the next big
platform startup!





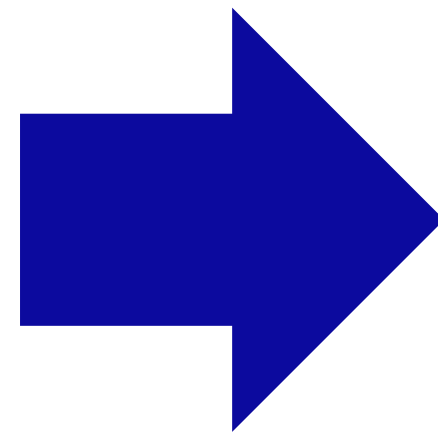
Dataset Intro



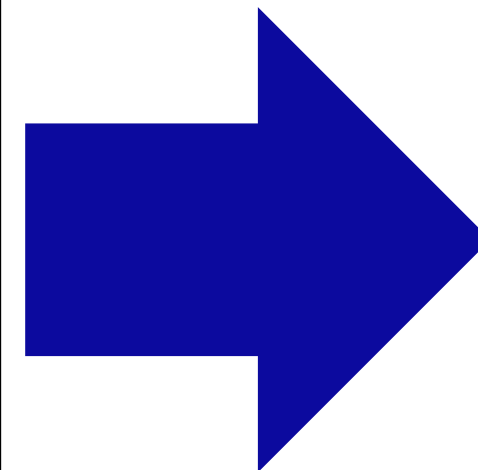
Dataset Creation Pipeline:



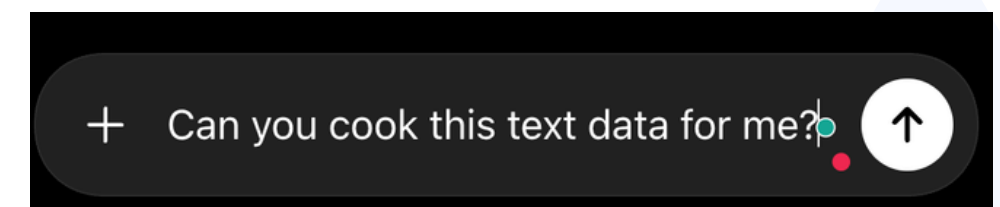
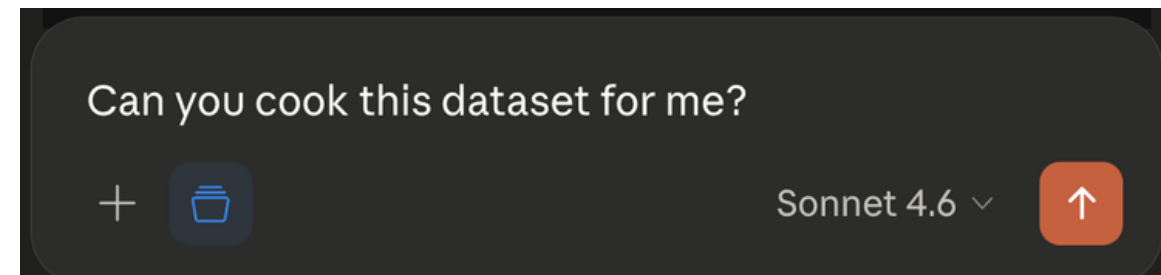
Took Startup Datasets from Kaggle.

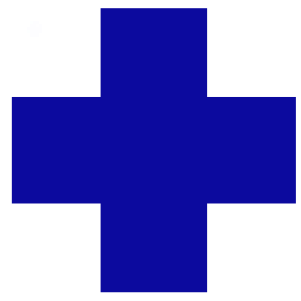


Told Claude to only merge and keep structured data on the platforms.



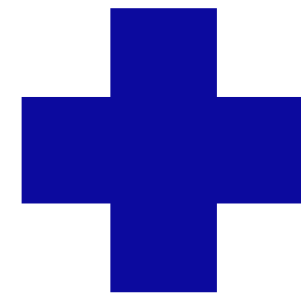
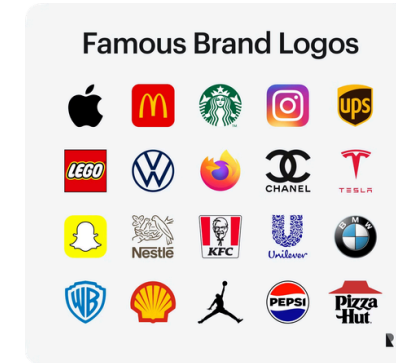
A. Used a Webscraper to build the textual columns.
OR
B. Autofilled text using ChatGPT.





Datasets Tested:

FILL IN THE (blank)
production



Wayback Machine Dataset

- Used Claude **web scraper** + Claude's **knowledge base**.
- **Founder** background sparse (100 rows) — **dropped**.
- 1090 Rows After null values.

ChatGPT Dataset

- Used **ChatGPT** to fill unstructured columns in a **template** format:
 - *"The company is a {platform_type} connecting two parties together."*
- 6877 rows After null values.

Logo Images

- Retrieved via Claude **web scraper**.
- A **3rd unstructured column**, merged with the previous two to create **new Datasets**.
- Wayback Machine Merged 695 rows
- ChatGPT Merged Dataset: 3893 rows.

Dataset Columns:

Structured Columns:

- Company Name
- Platform Type
- Seller Type
- Founded year
- First Funding Year
- Country
- State
- has_angel_investor
- has_seed_funding
- has_grant
- has_accelerator

Unstructured Columns:

- **About_text(In all Datasets)**
 - Description of company operations.
- **Founder Background(Only in Chatgpt):**
 - Description of the Founder's background.
- **Logo Images(in Separate Datasets for both Wayback and Chatgpt):**
 - Images of the company Logos.

Outcome Columns:

- **outcome(Binary):**
 - Success or Failure.
- **outcome_label(multiclass):**
 - IPO, Acquired, or Closed



Model Testing & Discussion

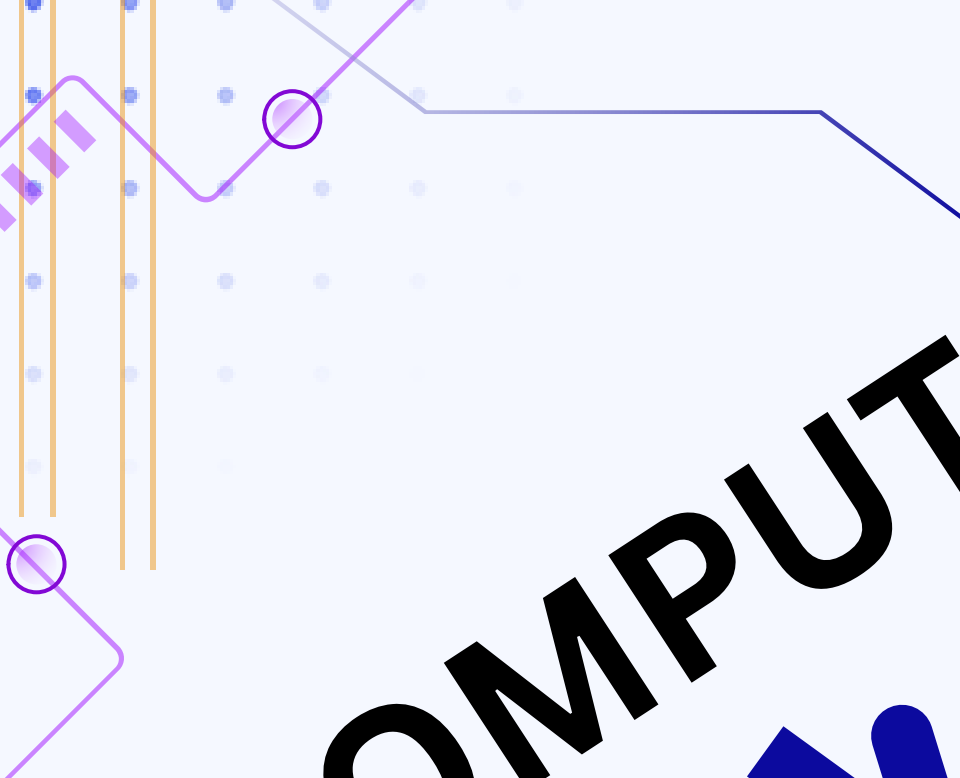


COMPUTE



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Phase 1: The Aspiration to Cook



Model Tested & Reasoning:

The Models:

1. Simple (Multi-Hot Vectorization)
2. Random Init Embeddings
3. RNNs:
 - Simple
 - LSTM
 - GRU
 - Their bidirectional versions
4. QKV Self-Attention
5. Multi-Head Attention

Reasoning:

- Covered in class.
- Deepen my code understanding.
- Accurately verify the code given by Claude

**Only the best models in
each Binary and
Multiclass go to Phase 2**

Model Testing Methodology:

Step 1:

Data Split: 60%
Train | 20%
Validation | 20%
Test — Stratified
& Shuffled

Step 2:

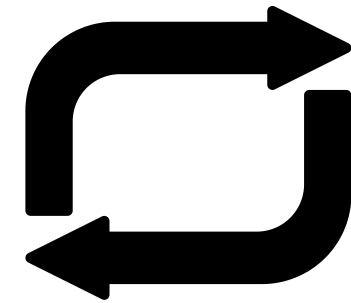
Binary Performance
Classes: Success |
Failure
Loss Function: Binary
Cross-Entropy

Step 3:

Multiclass Performance
Classes: IPO | Acquired | Closed
Class Weights Applied (IPO
imbalance)
Loss Function: Sparse Categorical
Cross-Entropy

Step 4:

Comparing The Results
Against each Respective
Benchmark and other
Models



For Dataset:
For Model :
Repeat Steps 2 and 3

Performance Metric:

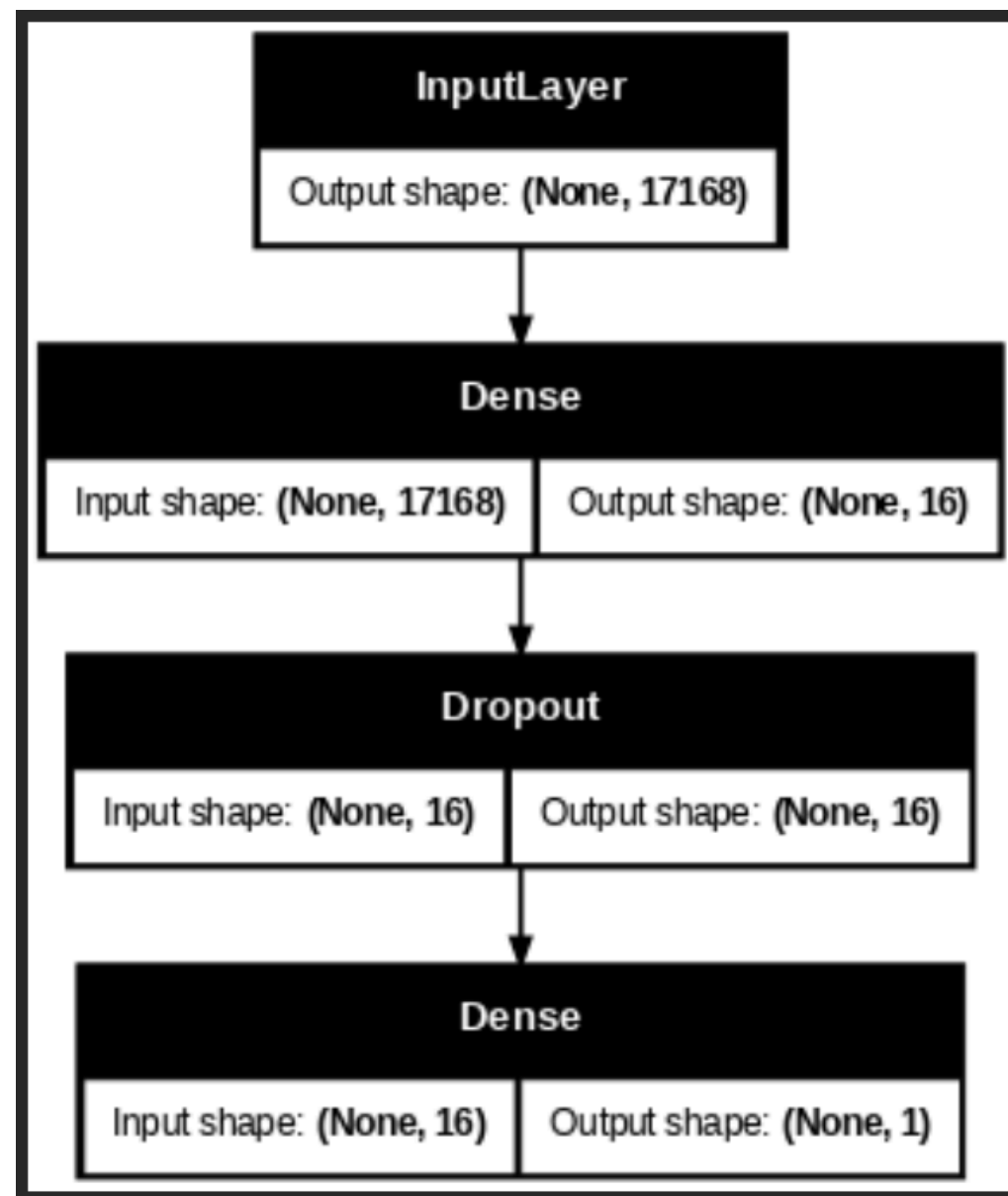
Accuracy

Reasons:

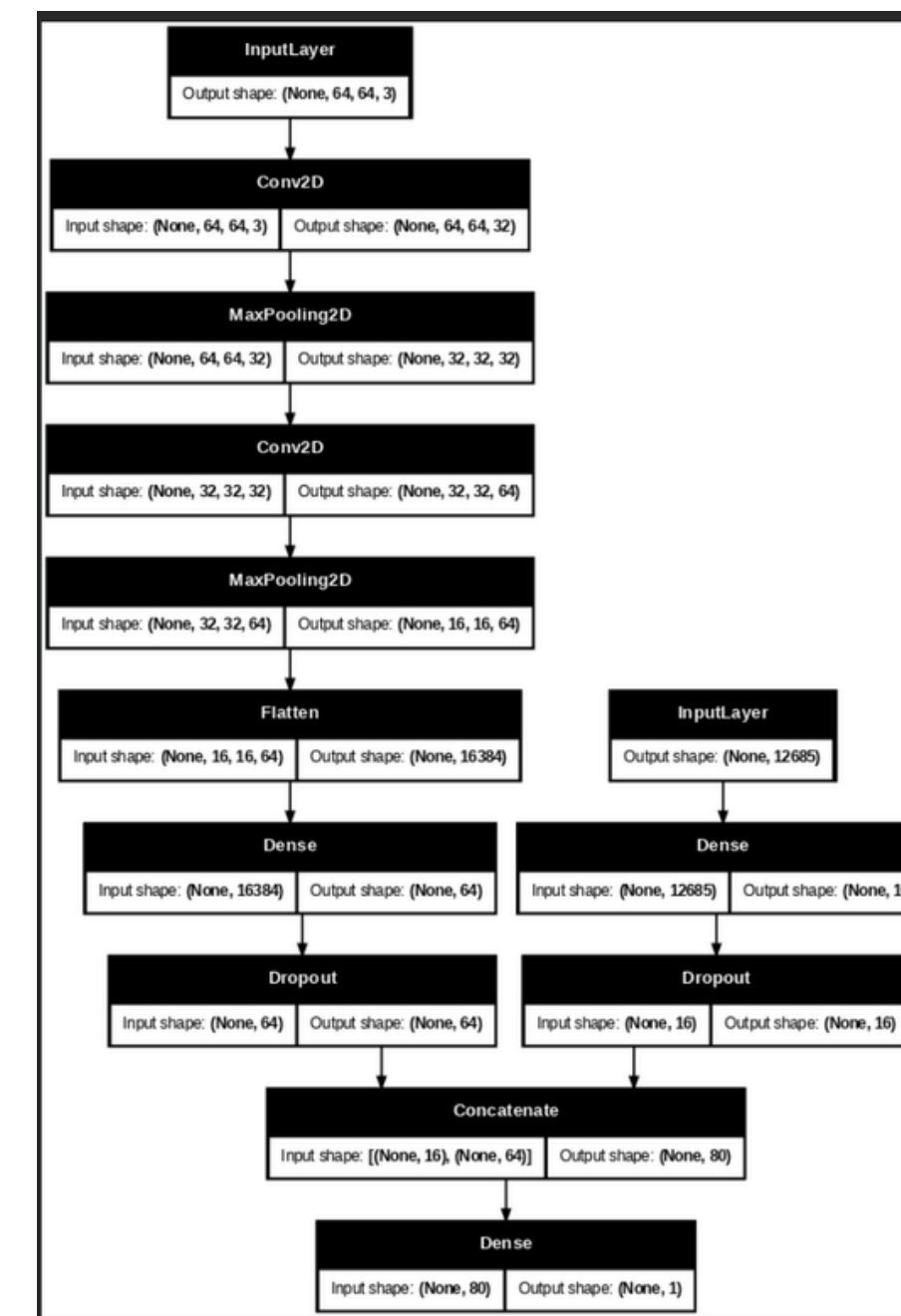
- Easy, Balanced Metric.
- VC firms may want a balance of Success/Failure Prediction.
- Easily Retrievable from Neural Networks.

Simple Model Architectures:

Simple Model In Wayback Machine and ChatGPT Datasets:

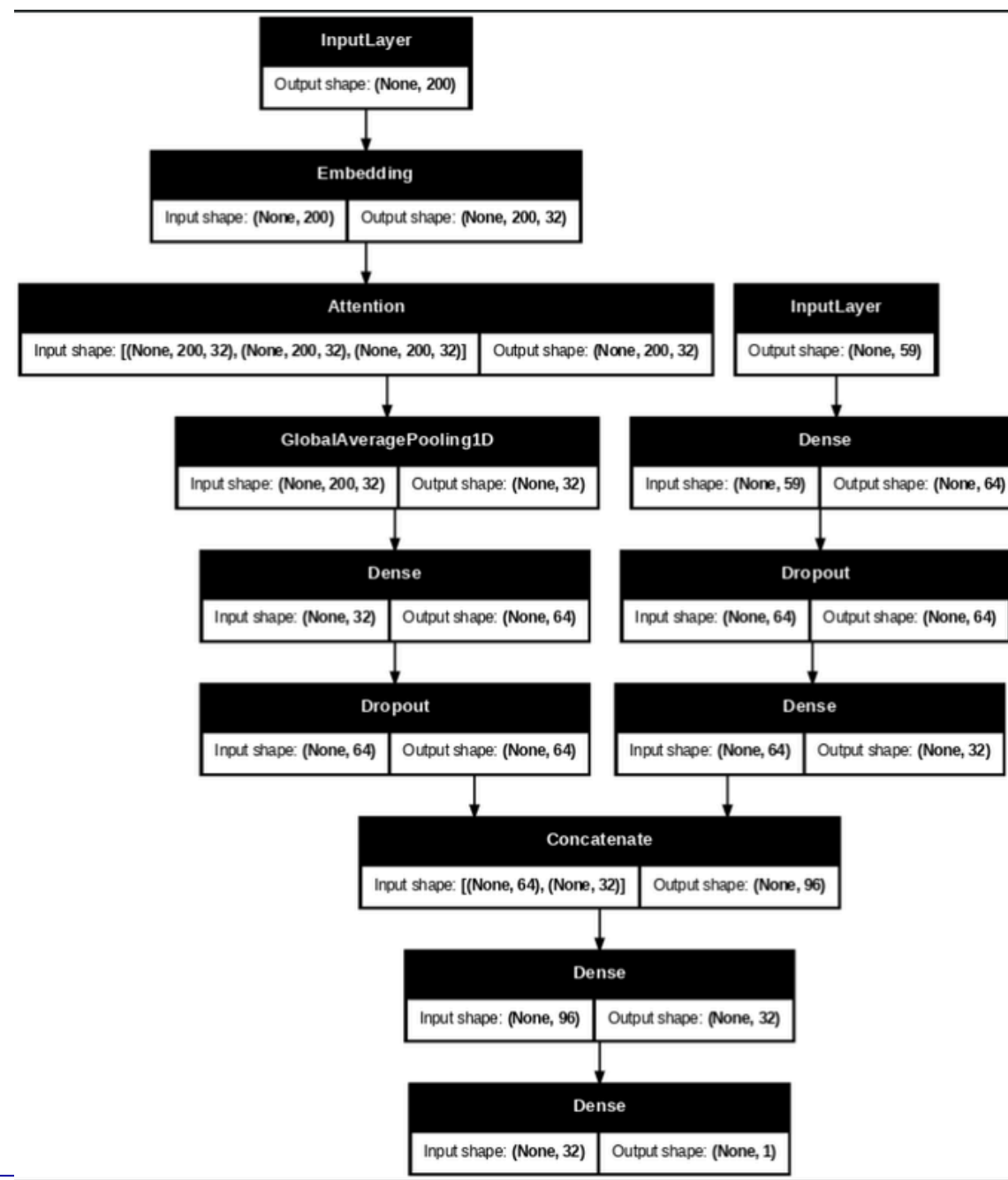


Simple Model In Image Wayback Machine and ChatGPT Datasets:

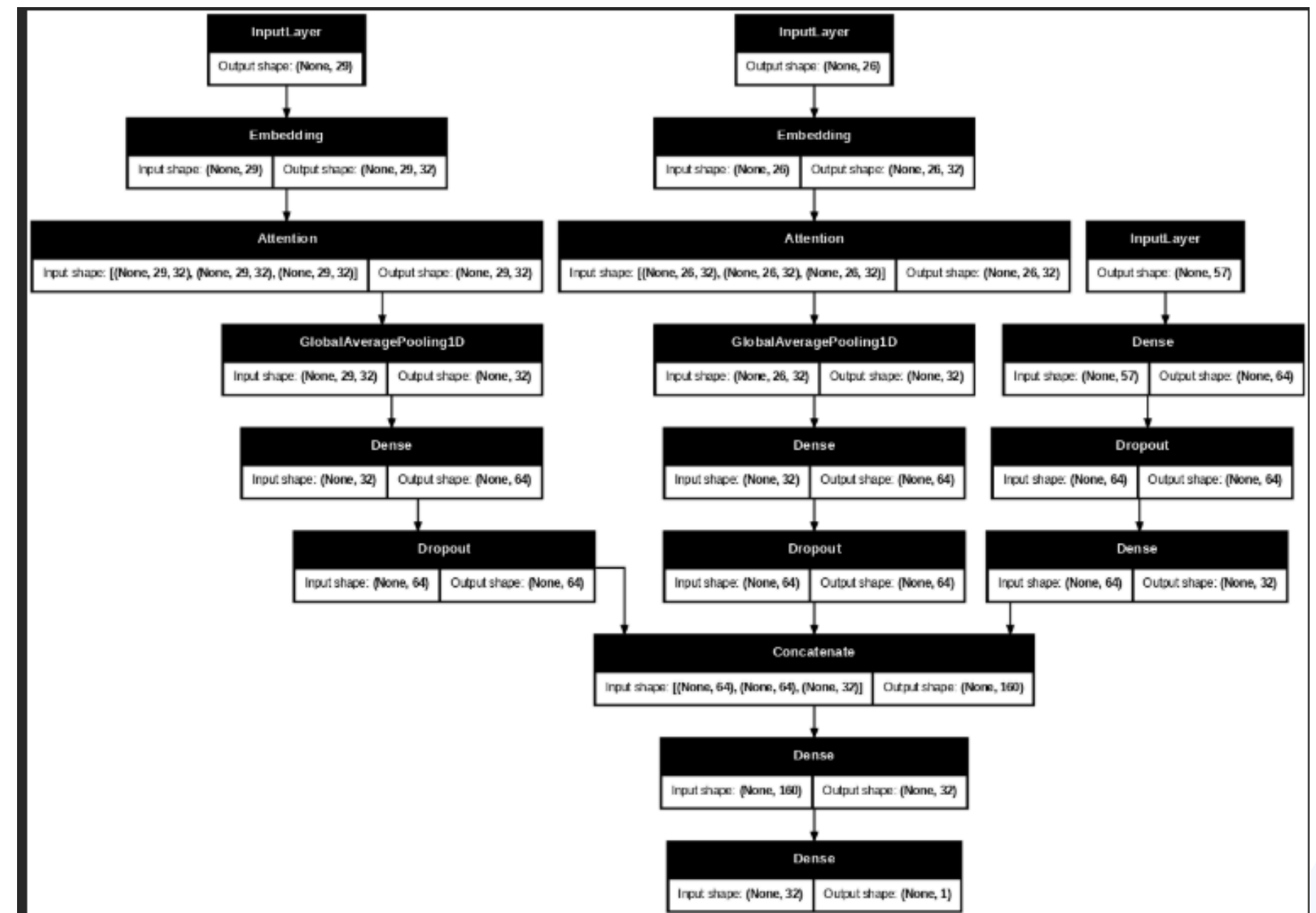


Most Model Architectures:

Most Models in the Wayback Machine Dataset:

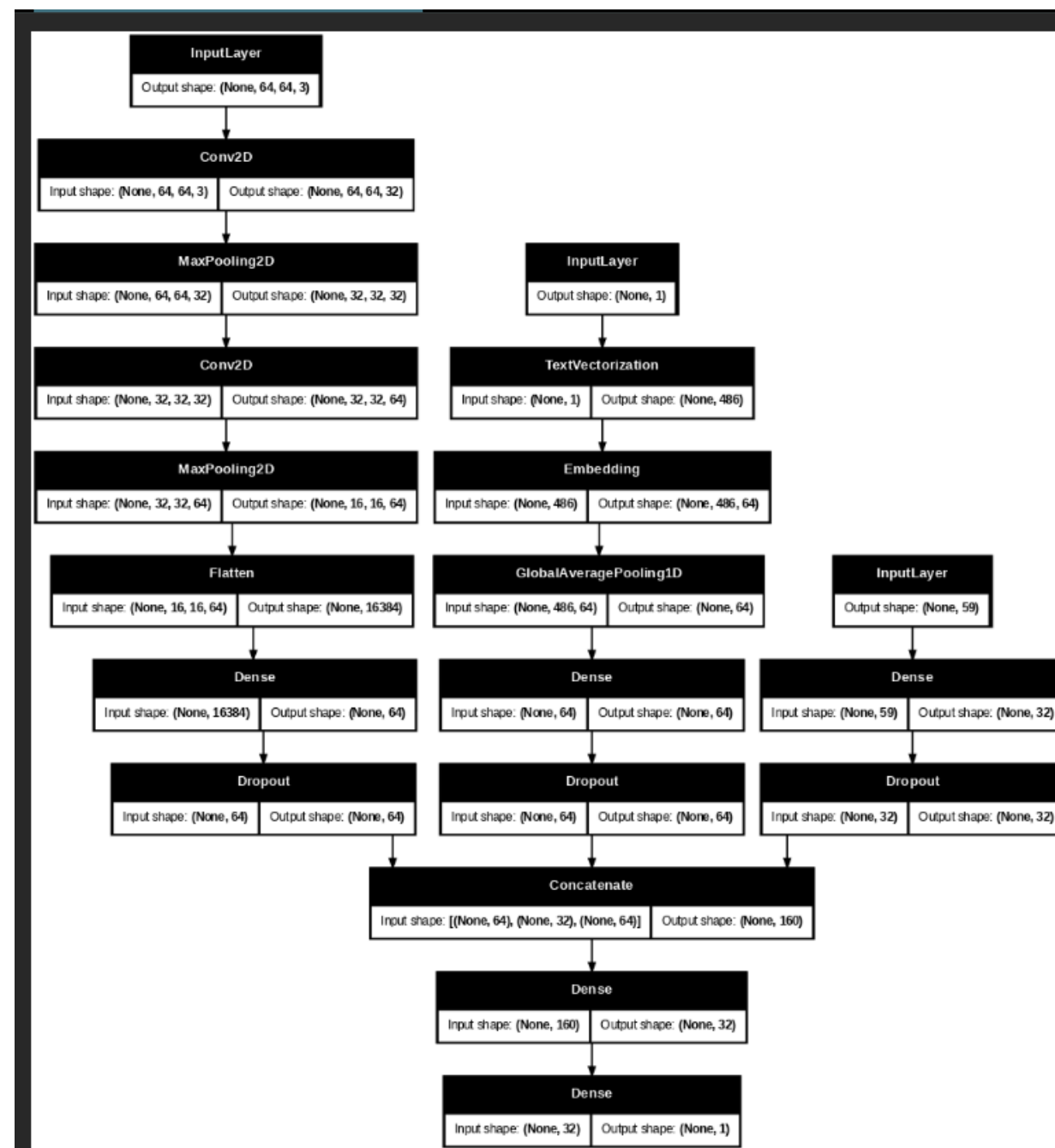


Most Models in the ChatGPT Dataset:

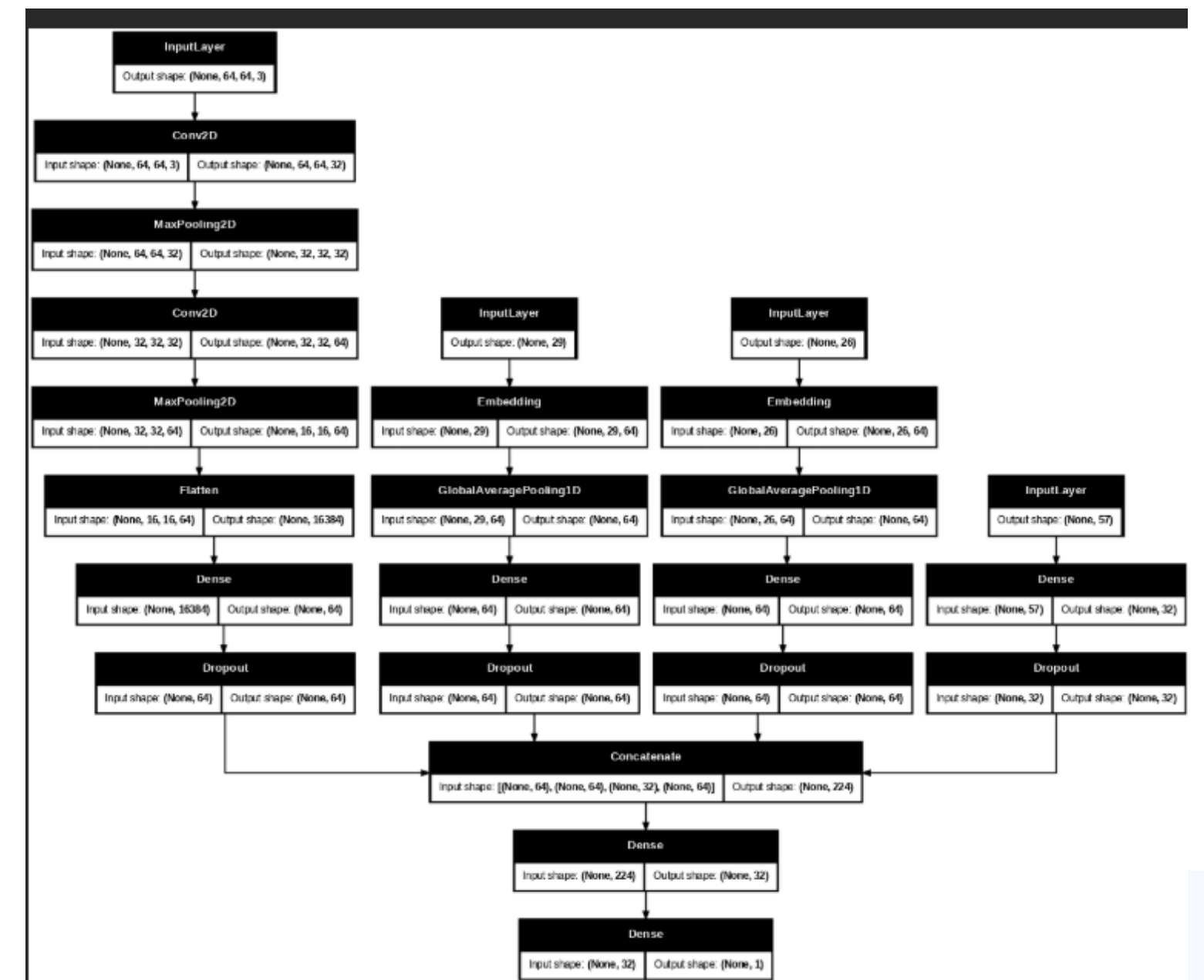


Most Image Model Architectures:

Most Models in the WayBack Machine Merged Dataset:

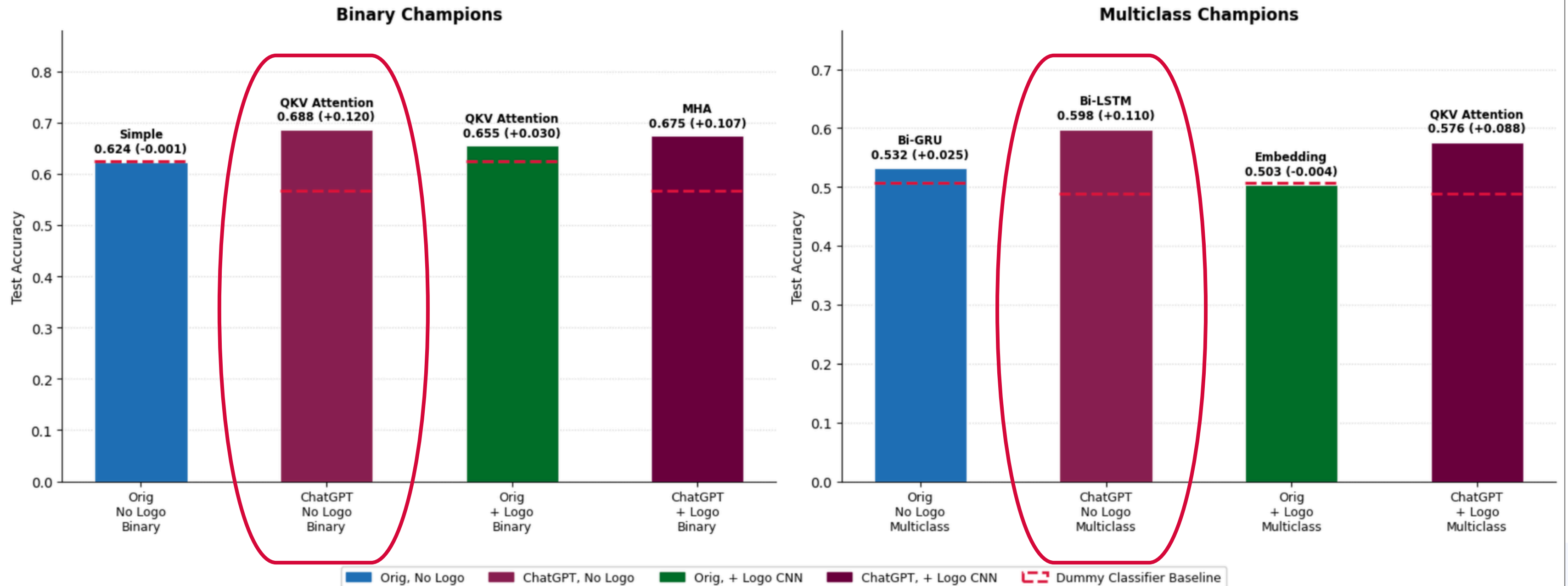


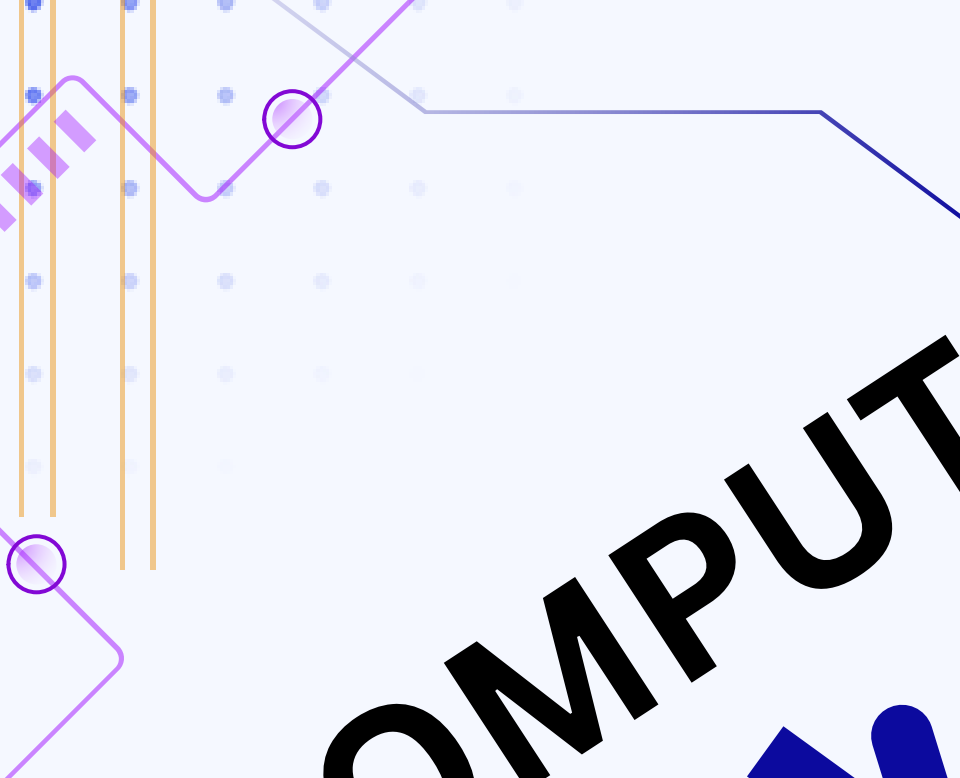
Most Models in the ChatGPT Machine Merged Dataset:



The Compute Games Final Results:

Final Champion Comparison — Best Model from Each Group





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Phase 2: The Rubric Assault



Embeddings Tested & Reasoning:

Embeddings:

1. Random Init Embeddings(Same as Phase 1).
2. Word2Vec Embeddings.
3. RoBERTa Embeddings.
4. FinBERT
5. SEC-BERT
6. FinancialBERT

Reasoning:

- Random Init serves as a baseline.
- Word2Vec and RoBERTa are rich in general language.
- FinBERT, SEC-BERT, and FinancialBERT capture the financial language more and may fit better for this project.

Cross-Validation Testing Methodology:

Step 1:

Data Split: 80%
Validation Pool |
20% Test —
Stratified &
Shuffled

Step 2:

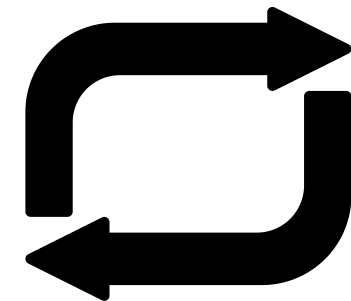
Encode the Textual
Columns Before with
the Embeddings.

Step 3:

Input into a run_cv function to
get CV results and use
plot_results to create visuals.

Step 4:

Comparing The Results
Against each Respective
Benchmark and other
Models



For Embedding:
For Model in (binary, multiclass) :
Repeat Steps 2 and 3

Performance Metrics:

- **Accuracy**
- **Recall**

Reasonings:

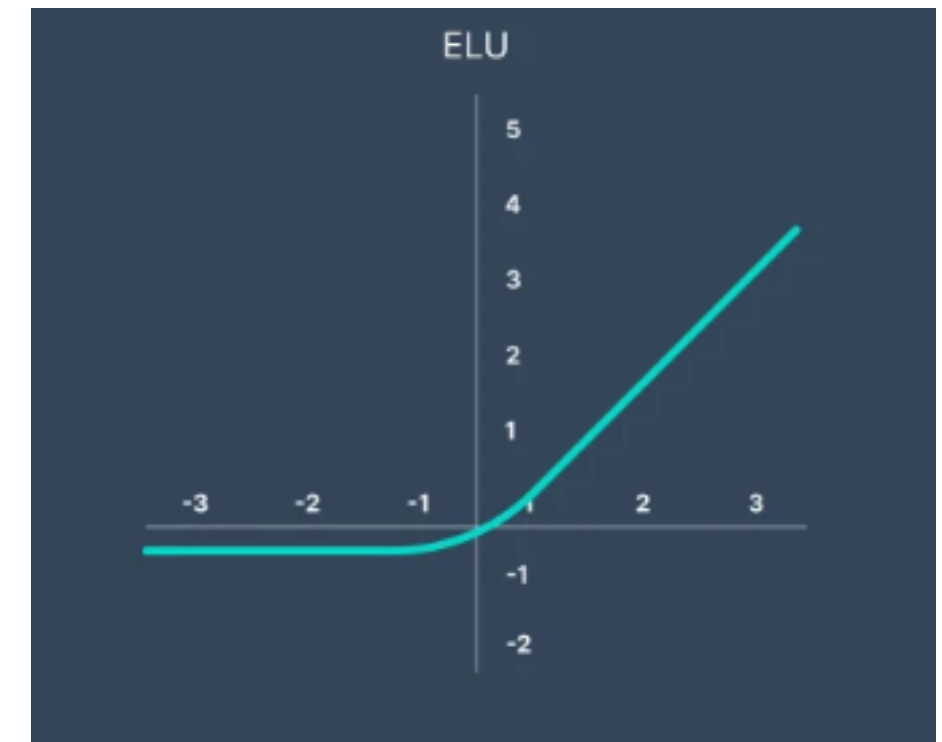
- Quick, balanced metric.
- VC firms want the model to capture the truest successes and failures.

Cross-Validation Sample Configurations:

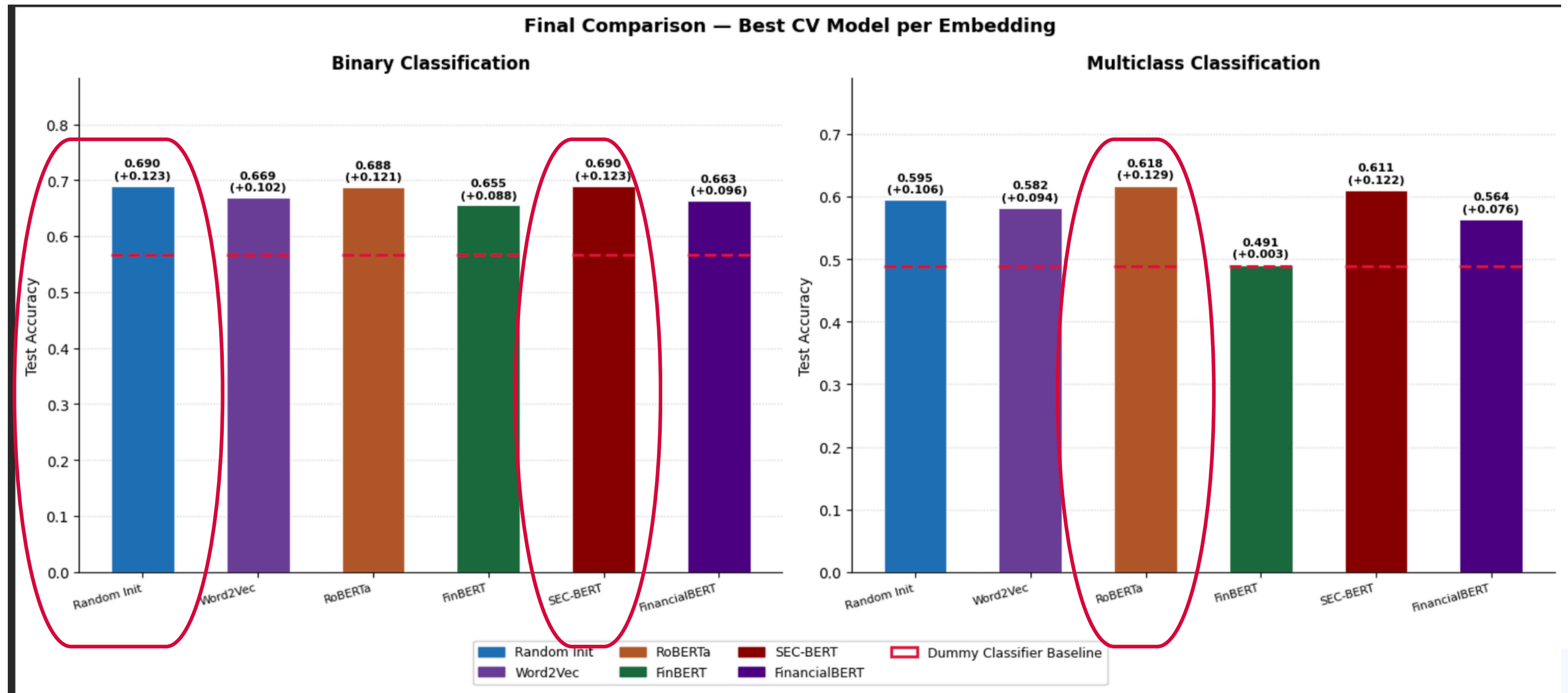
Parameters not covered in class:

```
SAMPLED_CONFIGS = [  
    (32, 1, 0.2, 0.001, 16, 'relu', 'rmsprop'),  
    (64, 1, 0.4, 0.0003, 32, 'elu', 'adam'),  
    (32, 2, 0.2, 0.0003, 16, 'elu', 'rmsprop'),  
    (64, 2, 0.4, 0.001, 32, 'relu', 'adam'),  
    (128, 1, 0.2, 0.001, 16, 'relu', 'adamw'),  
    (64, 2, 0.2, 0.0003, 16, 'elu', 'adamw'),  
    (32, 1, 0.4, 0.001, 32, 'relu', 'nadam'),  
    (128, 2, 0.4, 0.0003, 32, 'elu', 'nadam'),  
    (64, 1, 0.2, 0.001, 32, 'elu', 'adamw'),  
    (128, 1, 0.4, 0.0003, 16, 'relu', 'adam'),  
]
```

- **ELU:**
 - For positive values, functions like RELU.
 - For negative values, it smooths toward -1, solving the dead neuron problem.
- **ADAMW:**
 - Functions like ADAM, but gives weight decay after every update.



Cross-Validation Model Results:



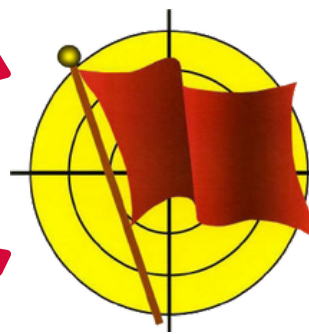
Cross-Validation Model Results:

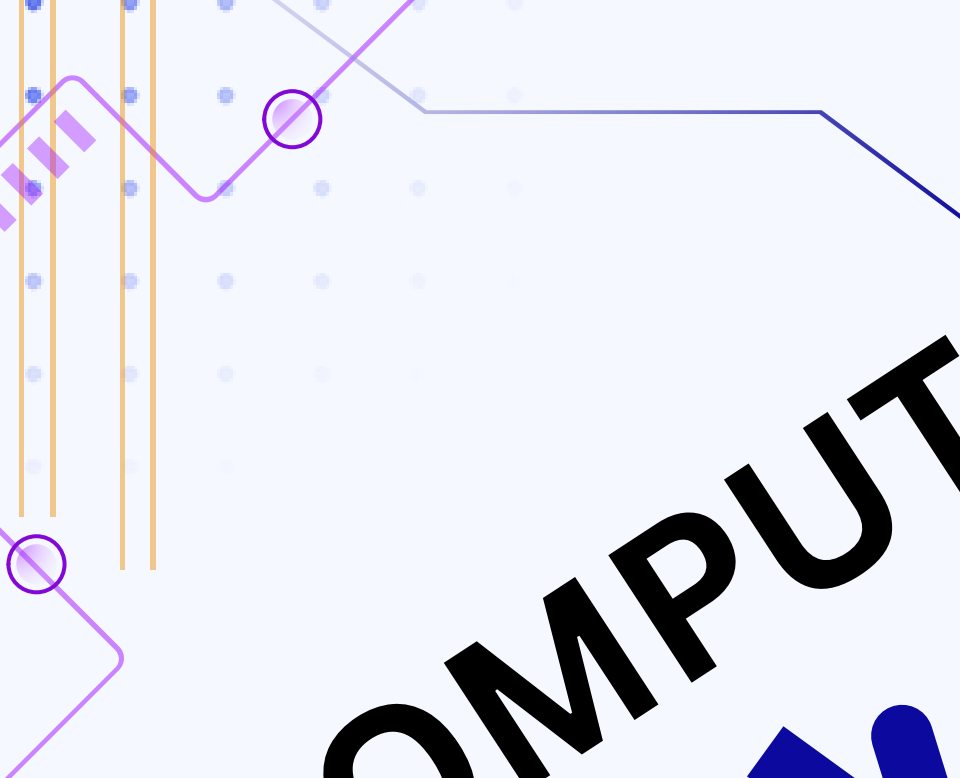
Binary Classification — Recall

Model	Failure	Success
Random Init	0.61	0.75
Word2Vec	0.50	0.80
RoBERTa	0.69	0.69
FinBERT	0.42	0.84
SEC-BERT	0.67	0.70
FinancialBERT	0.37	0.88

Multiclass Classification — Recall

Model	Closed	Acquired	IPO
Random Init	0.49	0.75	0.22
Word2Vec	0.46	0.72	0.43
RoBERTa	0.67	0.65	0.11
FinBERT	0.00	1.00	0.04
SEC-BERT	0.47	0.78	0.35
FinancialBERT	0.23	0.93	0.16





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Phase 3: Touching Grass & Reflecting



Project Improvements/Adjustments:

Data

Get Access to more and better data, possibly manually filling in the data using wayback machine.

Test Model out on current Startups

Interesting evaluation method, would be compared to results in sites like Pitchbook.

Try more models/architectures

Test out models that work for smaller datasets and with text columns.



QUESTIONS?

